

By the power of Grayskull

Small sample statistical power in Information Retrieval evaluation

Laurence A. F. Park Glenn Stone
Centre for Research in Mathematics
School of Computing, Engineering and Mathematics
University of Western Sydney, Australia
{l.park,g.stone}@uws.edu.au

ABSTRACT

Information Retrieval evaluation is typically performed using a sample of queries and a statistical hypothesis test is used to make inferences about the systems accuracy on the population of queries. Research has shown that the t test is one of a set of tests that provides the greatest statistical power while maintaining acceptable type I error rates, when evaluating with a large sample of queries. In this article, we investigate the effect of using a small query sample on the control of the type I error rate and change in type II error rate of a given set of hypothesis tests, meaning that the hypothesis tests may not satisfy Central Limit Theorem conditions. We found that all test performed similarly for unpaired tests. We also found that the bootstrap test provided greater power for the paired test, but violated the desired type I error rate for the smallest sample size (5 queries).

1. INTRODUCTION

Statistical hypothesis tests allow us to compute the probability of the state of a population parameter, based on a sample. For the case of evaluating the accuracy of an Information Retrieval system, we are assessing if the population mean score of the new system μ_B is greater than the population mean score of a baseline system μ_A using a sample from a given population of queries. The power of the test is the probability of it concluding that $\mu_B > \mu_A$, when μ_B really is greater than μ_A , for a given controlled type I error rate.

When we have a large query sample, we can use the Central Limit Theorem to identify the approximate distribution of the hypothesis test statistic, allowing us to control the type I error. If the query sample is small, we are unable to determine the distribution of the test statistic and hence we have less control of the type I error rate. Therefore, it may be that a test that provides greater power may be doing so by increasing the type I error.

In this article, we examine the statistical power of four hypothesis test types, taking note of the associated type I error rate, when using a small sample of queries. Just as He-Man receives the ultimate power from Castle Grayskull¹ and uses it for good, so too do we want to use statistical significance results for good, in the quest

for truth (hence the title of the article). We will identify the significance test that provides us with the greatest power, while maintaining the controlled type I error rate.

The contributions of this article is an analysis of the type I and II error rates of the t test, Wilcoxon test, permutation test and bootstrap test for when used for Information Retrieval evaluation with a small sample of queries.

The article proceeds as follows: We begin in Section 2 by describing the notion of statistical error and power and how it can be controlled by correctly modelling the test statistic. Section 3 examines previous investigations into the suitability of hypothesis tests for Information Retrieval evaluation. Finally, Section 4 contains an empirical analysis of the statistical power for each of the four test types.

2. INCREASING STATISTICAL POWER

Statistical Hypothesis testing is a method of determining the likelihood of two mutually exclusive hypotheses concerning a given population parameter. Performing the test requires:

1. Defining the Null and Alternative hypotheses (H_0 and H_A).
2. Gathering data as evidence.
3. Determining the probability of the evidence given the Null Hypothesis.

After completing a Hypothesis test, we arrive at decision; either we do not have enough evidence to reject H_0 and cannot assume H_A is true, or we have sufficient evidence to reject H_0 and therefore assume that H_A is true. Note that not rejecting H_0 does not mean that H_0 is definitely true, it only means that the evidence we gathered was not conclusive enough to reject it. Also, rejecting H_0 does not imply that H_0 is definitely false, it tells us that there is a low probability of obtaining the evidence we have gathered, given H_0 . Low probability does not mean no chance.

The typical hypothesis test used for Information Retrieval evaluation is as follows. Given two systems A and B, with evaluation score distributions Δ_A and Δ_B for a given population of queries, we want to test if the unknown mean system B score μ_B is greater than the unknown mean system A score μ_A . Therefore the Null Hypothesis is that both system mean scores are equal, giving us the two hypotheses:²

$$H_0 : \mu_A = \mu_B \text{ and } H_A : \mu_A < \mu_B \quad (1)$$

To test these hypotheses, we take a random sample A and B from the score distributions Δ_A and Δ_B and use the distribution of the observations in A and B to assess the probability of $\mu_A = \mu_B$. The

²We could also use $H_A : \mu_A \neq \mu_B$ to test for a difference, but the data we use later has no systems with equal means, meaning we would not be able to measure the type I error rate.

¹http://en.wikipedia.org/wiki/Castle_Grayskull

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.

ADCS '14 November 27 - 28 2014, Melbourne, VIC, Australia

Copyright is held by the owner/author(s). Publication rights licensed to ACM.

ACM 978-1-4503-3000-8/14/11...\$15.00

<http://dx.doi.org/10.1145/2682862.2682878>.

Truth	Test Outcome	
	Do not reject H_0	Reject H_0
$H_0 : \mu_B \leq \mu_A$	Correct	Type I error
$H_A : \mu_B > \mu_A$	Type II error	Correct

Table 1: The four possible outcomes from the hypotheses in Equation 1.

four outcomes of our assessment are shown in Table 1. Of the four outcomes, there are two correct outcomes: the first where the Null Hypothesis is not rejected and the true value of μ_A is greater than or equal to μ_B , the second where the Null Hypothesis is rejected and the true value of μ_B is greater than the true value of μ_A . The two remaining outcomes are errors: a type I error being when the Null Hypothesis is rejected when it was true, and a type II error being when the Null Hypothesis is not rejected when it should have been. The probability of type I and II errors are given by the type I and II error rates. The power of the test is 1 - type II error rate.

For a given statistical test, we want to control the type I error rate and maximise the power (minimise the type II error rate) to ensure that the correct outcome is reached. Figures 1 and 2 show the probability of a type I and type II error, given a Normally distributed test statistic. Figure 1 provides the case where the Null Hypothesis is true. We see that most values of the test statistic do not reject the Null Hypothesis (the correct outcome). We reject the Null Hypothesis when the test statistic is unusually large, leading us to believe that the Alternative Hypothesis is true, when it is not.

If we know the distribution of the test statistic under the Null Hypothesis, we can control the type I error by setting a critical value. When the test statistic is greater than this critical value, we reject H_0 . The type I error is then computed as the probability of the test statistic exceeding this critical value by chance, causing us to incorrectly reject H_0 . Rather than comparing the test statistic to the critical value, we map these two values to probabilities, allowing us to compare a significance level α to the test's p value. For example, if $\alpha = 0.05$, we reject H_0 when the test's p value is less than 0.05, meaning that the type I error rate is equal to α . Therefore, by setting a value for α , we also determine the type I error rate.

Figure 2 depicts the case where the mean of the test statistic is greater than the assumed mean under the Null Hypothesis. The figure shows that the correct outcome is achieved when the test statistic is greater than the critical value; note that the probability of this correct outcome is the power of the test. Unfortunately, the Null Hypothesis is not rejected when the test statistic is not greater than the critical value, giving a probability of error shown by the shaded region of the test statistic true distribution. To measure the type II error, given by the shaded region of Figure 2, we must know the value of $\mu_B - \mu_A$. Note that the point of the hypothesis test is to determine if $\mu_B - \mu_A > 0$, therefore if $\mu_B - \mu_A$ is known there is no need for the test. This means that $\mu_B - \mu_A$ is not known during the test time, making it difficult to measure the type II error rate and power.

From Figures 1 and 2, we can see that if the critical value is shifted to the right, we reduce the type I error rate, but also reduce the power of the test. The critical value c is determined by α and the distribution of the test statistic $\bar{b} - \bar{a}$ under the Null Hypothesis:

$$P(\bar{b} - \bar{a} > c | H_0) = \alpha \quad (2)$$

therefore if we poorly estimate the test statistic distribution, we will obtain an inaccurate c and hence the type I error rate will not be α . If it happens that poorly estimated test statistic distribution leads to the true type I error rate being less than α , the power of the test will

also be reduced. On the other hand, if the true type I error rate is greater than α , the power of the test is greater, but a significant difference will be falsely declared for more system pairs. Therefore, for fair system comparisons, we must ensure that the type I error rate of a test be close to the assigned α .

3. RELATED WORK

Smucker et al. [3] examined the correlation between the t test, permutation test, bootstrap test, signed rank test and sign test for samples of 50 queries and then later using 10, 20, 30, 40 and 50 queries with the t test, permutation test and bootstrap test [4]. Each of the experiments were performed on TREC 3 and TREC 5-8 using average precision. The results showed that the t test, permutation test and bootstrap test provided similar p values for 50 queries and that as the number of queries reduces, the similarity reduces. They discuss the assumptions behind the five tests and conclude that permutation test is the most trustworthy; since the signed rank test and sign test were the least similar to the permutation test, they conclude that they should not be used for information retrieval evaluation.

Urbano et al. [5] compared the same five tests using samples of the average precision from 50 queries within the TREC 2004 Robust collection. The experiment involved randomly sampling mutually exclusive pairs of 50 queries and comparing the results of the given test. The results showed that the sign test performed poorly, while the remaining four tests provided similar results. The t test was the most consistent test, while all tests provided estimated type I error that was closely aligned to the true type I error.

When using 50 queries each of these investigations concluded that the t test was either the preferred or equal preferred test from the set of five tests. This result is not surprising since each of the experiments involved random sampling from some population, where the random sample was large, implying that the Central Limit Theorem would be evident. Therefore the distribution of the mean difference in average precision would be approximately Normal, implying that the t test assumptions would be valid.

Small sample experiments by Park [1, 2] showed that the distribution of sample mean average precision is approximately Normal, but unfortunately the ratio of the sample and population standard deviations is not proportional to χ^2 , meaning that the sample statistic will not be t distributed for small query sample sizes.

In this article we will examine the small sample case of the type I and II error rates and hence the statistical power. The assumptions behind the Central Limit Theorem will be violated, therefore we expect that the difference between each of the tests will be more obvious. We will examine the four tests used in the above previous work, apart from the sign test (since each previous investigation showed it to be inferior to the other four tests).

4. EVALUATION OF HYPOTHESIS TESTS

In this section, we will examine each of the four types of hypothesis tests (t test, Wilcoxon test, permutation test, and bootstrap test) and their effect on the type I and II error rate for evaluation using varying query sizes.

To compute the type I and type II error rate for a test, we must be able to randomly sample a set of scores from a pair of system's score distributions Δ_A and Δ_B . Unfortunately, we do not know the functional form of the system distributions, but we are able to approximate them using an empirical distribution from a large sample of scores. The TREC 2004 Robust track, containing 249 queries, was used by Urbano et al. [5] and is the largest collection that could be obtained with relevance judgements. We obtained the Average Precision (AP) and Standardised Average Precision (sAP)

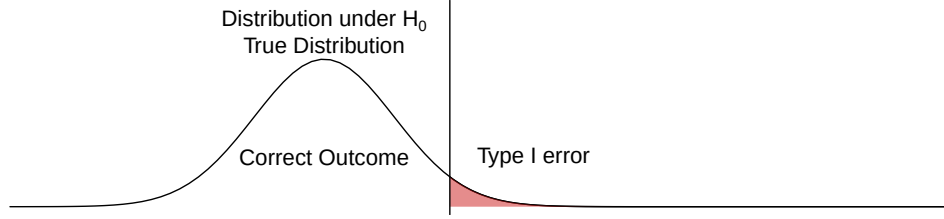


Figure 1: An example of the distribution of a test statistic (usually $\bar{b} - \bar{a}$) under the assumption of the Null Hypothesis and the outcome when the Null Hypothesis is true. The vertical line shows the critical value; when the test statistic is greater than this critical value, we mistakenly reject the Null Hypothesis (type I error). The shaded region shows the probability of a type I error.

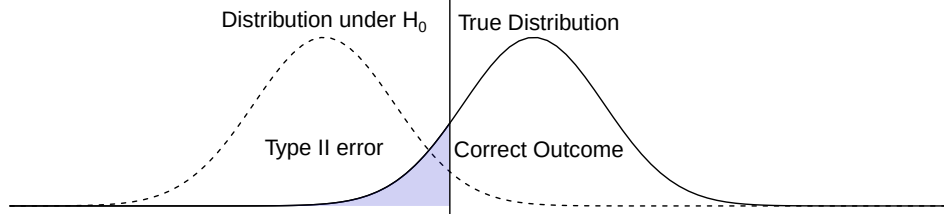


Figure 2: An example of the distribution of a test statistic (usually $\bar{b} - \bar{a}$) under the assumption of the Null Hypothesis (the dotted distribution), the true distribution of the test statistic, and the outcome when the Null Hypothesis is false. The vertical line shows the critical value; when the test statistic is less than this critical value, we mistakenly do not reject the Null Hypothesis (type II error). The shaded region shows the probability of a type II error. Note that the power of the test is the probability of the test statistic appearing to the right of the critical value.

[6] evaluation scores for a set of 110 systems over the 249 queries, providing us with an estimate of the AP and sAP distribution for each of the systems. Note that the TREC queries are constructed, and so are not a *random* sample from a general query pool, but can be considered a random sample from a test query pool, where the queries are designed to discriminate systems. Therefore the mean score from the 249 queries is a good estimate of the population mean score computed from the test query distribution.

The paired experiment was run for each of the four hypothesis tests by sampling two systems from the pool of 110, then sampling a set of n queries and the associated scores for each system, without replacement. The p value of the paired test was computed and compared to α to determine the outcome of the test. The outcome of the test was then compared to an estimated *truth*, where the truth was estimated by comparing the means using the entire set of 249 queries. This process was repeated 10,000 times to obtain the type I and II error rates. The unpaired experiment proceeded just as the paired experiment, except that two mutually exclusive sets of n queries were sampled from the query pool. The evaluation scores for the first system were produced for the first set of n queries, while the evaluation scores for the second system were produced for the second set of n queries. For each permutation and bootstrap test, we used samples of size 1,000 to obtain the permutation and bootstrap distributions. To examine the behaviour of the type I and II error rate for small query samples, we performed the experiments for samples of size $n = 5, 10, 15$ and 20 queries.

The results for the paired experiment using $\alpha = 0.05$ are shown in Tables 2 and 3. Table 2 shows the type I error rate for AP and sAP scores for each of the four tests. We set $\alpha = 0.05$, therefore we expect the type I error rate to be 0.05, but we find that the values drop to 0.01 as the query size increases, showing us that the tests are conservative. We also find that the permutation and bootstrap tests provide a type I error rate greater than α when $n = 5$, meaning

Paired Test (Type I)	Number of Queries			
	5	10	15	20
<i>AP Scores</i>				
<i>t</i> Test	0.039	0.017	0.011	0.011
Signed Rank Test	0.031	0.017	0.014	0.013
Permutation test	0.069	0.024	0.016	0.009
Bootstrap test	0.081	0.029	0.019	0.012
<i>Standardised AP Scores</i>				
<i>t</i> Test	0.046	0.016	0.014	0.006
Signed Rank Test	0.035	0.022	0.012	0.006
Permutation test	0.064	0.018	0.013	0.007
Bootstrap test	0.076	0.032	0.020	0.011

Table 2: The measured *Type I* error rate for each of the four types of *Paired Hypothesis tests* using either AP Scores or Standardised AP Scores, where $\alpha = 0.05$.

that these tests should have greater power for $n = 5$ since the type I error is greater than desired. Otherwise, all error rates are similar for each n .

Table 3 shows the type II error rate for AP and sAP across all four tests. We can see that the error rate decreases as the number of queries increases and the *t* test and Signed Rank test provide very similar error rates. As expected, the permutation and bootstrap tests have lower type II error for $n = 5$, and we can see that this continues for the bootstrap test for the remaining query sizes.

The results for the unpaired experiment using $\alpha = 0.05$ are shown in Tables 4 and 5. Table 4 shows the type I error rate for AP and sAP for each of the four tests. We find that each of the test results are similar for a given number of queries and that they reduce as the number of queries increases. We find that the type I

Paired Test (Type II)	Number of Queries			
	5	10	15	20
<i>AP Scores</i>				
<i>t</i> test	0.446	0.383	0.356	0.325
Signed Rank Test	0.445	0.388	0.353	0.329
Permutation test	0.422	0.386	0.353	0.327
Bootstrap test	0.388	0.348	0.338	0.316
<i>Standardised AP Scores</i>				
<i>t</i> test	0.444	0.376	0.348	0.336
Signed Rank Test	0.448	0.387	0.347	0.317
Permutation test	0.428	0.371	0.349	0.334
Bootstrap test	0.377	0.344	0.332	0.312

Table 3: The measured *Type II* error rate for each of the four types of *Paired Hypothesis* tests using either AP Scores or Standardised AP Scores, where $\alpha = 0.05$.

Unpaired Test (Type I)	Number of Queries			
	5	10	15	20
<i>AP Scores</i>				
<i>t</i> test	0.112	0.073	0.039	0.032
Rank Sum Test	0.141	0.080	0.060	0.034
Permutation test	0.127	0.074	0.052	0.039
Bootstrap test	0.166	0.077	0.065	0.032
<i>Standardised AP Scores</i>				
<i>t</i> test	0.124	0.076	0.056	0.040
Rank Sum Test	0.125	0.069	0.051	0.043
Permutation test	0.121	0.067	0.054	0.038
Bootstrap test	0.175	0.096	0.059	0.036

Table 4: The measured *Type I* error rate for each of the four types of *Unpaired Hypothesis* tests using either AP Scores or Standardised AP Scores, where $\alpha = 0.05$.

error rate is greater than the set $\alpha = 0.05$ for 5 and 10 queries over all tests, and the type I error is similar to α for $n = 15$.

Table 5 shows the type II error rate for AP and sAP for each of the four tests. We find that the error rates are all very similar for each query size, with the Bootstrap test being slightly lower. We also see the error rate dropping as the query sample size increases, as expected.

All of the error rates for each test converged as n increased. This is likely due to the test statistic becoming more Normal as n increases (according to the Central Limit Theorem) and the tests computing similar critical values. The results show that there is little difference between the error rates for AP compared to sAP and that there is little difference between the unpaired tests over all samples sizes. We did observe that the Bootstrap test provided greater power for the paired tests. This is likely partially due to the type I error rate of the Bootstrap test being greater than the other tests. We also found that the bootstrap and permutation paired test type I error for $n = 5$ was greater than the wanted 0.05, but the *t* test and Signed Rank test were conservative for all samples sizes.

5. CONCLUSION

Statistical hypothesis tests allow us to make inferences about population parameters based on sample statistics. Information Retrieval evaluation is typically performed by comparing the response of two systems using a sample of queries with a defined document collection. Therefore, hypothesis tests are used to provide a confidence level of our belief that one system is more accurate than another.

Unpaired Test (Type II)	Number of Queries			
	5	10	15	20
<i>AP Scores</i>				
<i>t</i> test	0.462	0.440	0.430	0.413
Rank Sum Test	0.476	0.445	0.424	0.405
Permutation test	0.469	0.448	0.434	0.407
Bootstrap test	0.447	0.435	0.426	0.407
<i>Standardised AP Scores</i>				
<i>t</i> test	0.470	0.444	0.433	0.409
Rank Sum Test	0.458	0.450	0.430	0.410
Permutation test	0.467	0.451	0.419	0.407
Bootstrap test	0.454	0.442	0.421	0.402

Table 5: The measured *Type II* error rate for each of the four types of *Unpaired Hypothesis* tests using either AP Scores or Standardised AP Scores, where $\alpha = 0.05$.

Research has shown that the *t* test is sufficient for comparing the mean accuracy of two systems, when using a large query sample. This is likely due to the test statistic being approximately *t* distributed for large sample sizes. Unfortunately, we are unable to make this assumption about the distribution of the test statistic for small sample sizes.

In this article, we examined the type I and II error rates of the *t* test, Wilcoxon test, permutation test, and bootstrap test when evaluating using only a small number of queries. We found that all tests provided similar power and type I error for unpaired queries. We also found that the bootstrap test provided the greatest power of the four paired tests while maintaining an appropriate type I error rate for all cases except for sample sizes of 5. This is likely due to the other tests being more conservative in terms of type I error rate.

Acknowledgement The authors thank Falk Scholer for his comments and advice on Information Retrieval evaluation methods.

References

- [1] L. A. F. Park. Confidence intervals for information retrieval evaluation. In A. Turpin, F. Scholer, and A. Trotman, editors, *Proceedings of the Fifteenth Australasian Document Computing Symposium*, 2010. URL <http://staff.scm.uws.edu.au/~lapark/publications/confidence-2010.pdf>.
- [2] L. A. F. Park. Bootstrap confidence intervals for mean average precision. In *Proceedings of the Fourth ASEARC Conference*, pages 51–54, 2011. URL <http://staff.scm.uws.edu.au/~lapark/publications/bootstrap-ap.pdf>.
- [3] M. D. Smucker, J. Allan, and B. Carterette. A comparison of statistical significance tests for information retrieval evaluation. In *Proceedings of the sixteenth ACM conference on Conference on information and knowledge management*, pages 623–632. ACM, 2007.
- [4] M. D. Smucker, J. Allan, and B. Carterette. Agreement among statistical significance tests for information retrieval evaluation at varying sample sizes. In *Proceedings of the 32Nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '09, pages 630–631, New York, NY, USA, 2009. ACM. ISBN 978-1-60558-483-6. doi: 10.1145/1571941.1572050. URL <http://doi.acm.org/10.1145/1571941.1572050>.
- [5] J. Urbano, M. Marrero, and D. Martín. A comparison of the optimality of statistical significance tests for information retrieval evaluation. In *Proceedings of the 36th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '13, pages 925–928, New York, NY, USA, 2013. ACM. ISBN 978-1-4503-2034-4. doi: 10.1145/2484028.2484163. URL <http://doi.acm.org/10.1145/2484028.2484163>.
- [6] W. Webber, A. Moffat, and J. Zobel. Score standardization for inter-collection comparison of retrieval systems. In *Proceedings of the 31st annual international ACM SIGIR conference on Research and development in information retrieval*, pages 51–58. ACM, 2008.